For M/V XIANG RUI KOU only

Rev.	Alteration or remark	Date	Drawn	Appr.

DELTAMARIN	Drawn MVa / 04.07.2011	Scale	COSCO 50000 DWT SSHLV
	Checked PJA / 04.07.2011		
	Appvd. PJA / 04.07.2011	Metr.	
Name FINAL LOADING MANUAL ADDENDUM 2			Revision
			Page 2+52
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CONTENTS

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1 General

This document includes loading conditions for submerging condition. Loading conditions for aftload and sideload are also presented.

This document is an additional part for loading manual. The document does not include all relevant background information relating operation of the vessel in presented conditions. Fore that reason this addendum should always be used together with the loading manual.

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2 Submerged condition

2.1 Summary of loading conditions

CASE	TEXT	DISP	DWT	BW	CARGO	T	TR	HEEL	KMT
		t	t	t	t	m	m	degree	m
LC41	Submerging condition 26m, departure	109576.5	88705.1	84284.0	0.0	26.000	0.001	0.0	9.760
LC42	Submerging condition 26m, arrival	109576.2	88704.8	87440.6	0.0	26.000	0.000	0.0	9.760

CASE	KGF	GM	GMCORR	BMMIN	BMMAX	RELBM	XRELBM	SHMIN	SHMAX	RELSH	XRELSH
	m	m	m	tm	tm	%	m	t	t	%	m
LC41	8.99	0.774	0.132	-33003	28233	47.4	8	-1665	2057	33.0	199
LC42	8.89	0.866	0.116	-71777	31585	47.4	8	-2887	2547	33.3	82

CASE	name	
TEXT	descriptive name	
DISP	total displacement	t
DWT	mass of load	t
BW	mass of load	t
CARGO	mass of load	t
T	draught, moulded	m
TR	trim	m
HEEL	heeling angle	degree
KMT	transv. metac. height	m
KGF	cgz of load	m
GM	metacentric height	m
GMCORR	GM correction	m
BMMIN	minimum bending moment	tm
BMMAX	maximum bending moment	tm
RELBM	relative bending moment	%
XRELBM	x where max. relative bending mom.	m
SHMIN	minimum shear force	t
SHMAX	maximum shear force	t
RELSH	relative shear force	%
XRELSH	x where max. relative shear force	m

2.2 LC41 Submerging condition 26m, departure

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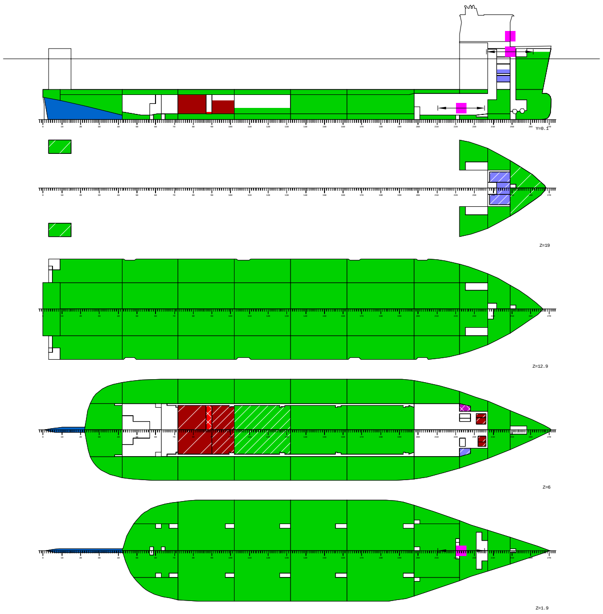


LOADING CONDITION LC41: Submerging condition 26m, departure

FLOATING POSITION

Mean draught (moulded)	26.00 m	KM above the moulded base	9.76 m
Draught at AP (moulded)	26.00 m	KG above the moulded base	8.85 m
Draught at FP (moulded)	26.00 m	GM0 (solid)	0.91 m
Trim	0.00 m	Free surface correction	-0.13 m
Heeling	0.0 deg		
		GM (fluid)	0.77 m
Mean draught (heel 0)	26.00 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	3500.0	71.26	-0.31	6.16	2300.7
Diesel Oil	80.0	152.31	1.43	3.22	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	20.0	179.80	-9.33	6.25	10.1
Fresh Water	300.0	195.55	0.00	18.24	343.4
Ballast Water	84284.0	113.41	0.01	8.54	11527.8
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	88705.1	111.89	0.00	8.48	14464.8
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	109576.5	112.18	-0.00	8.85	14464.8

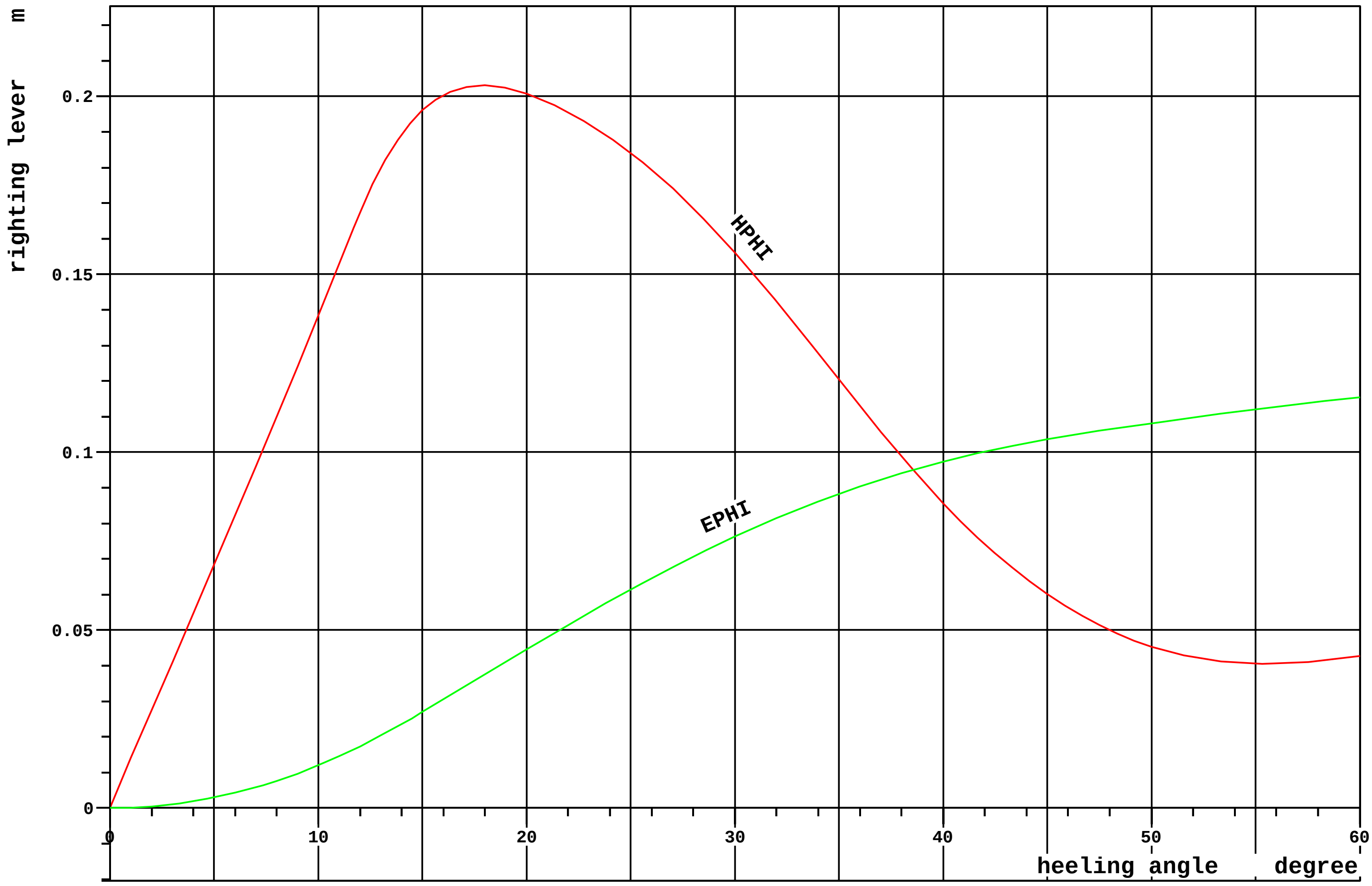
LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HFO1	HFO Storage Tank 1	1031.0	98.0	63.60	5.51	6.61	0.0
HFO2	HFO Storage Tank 2	1237.2	98.0	64.80	-5.51	6.61	0.0
HFO3	HFO Storage Tank 3	574.3	69.5	76.72	5.32	5.47	1143.5
HFO4	HFO Storage Tank 4	574.4	69.5	76.72	-5.32	5.47	1143.5
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		3500.0		71.26	-0.31	6.16	2300.7
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	20.1	10.5	70.80	5.69	3.02	259.6
T09.07	MDO Storage P	24.4	78.6	179.30	9.58	3.38	6.4
T09.08	MDO Storage S	24.4	78.6	179.30	-9.58	3.38	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		80.0		152.31	1.43	3.22	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.0	49.8	179.80	-9.33	6.25	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	80.0	36.5	194.80	4.81	17.46	60.2
FW2	Fresh Water S	80.0	36.5	194.80	-4.81	17.46	60.2
FW3	Fresh Water C	70.0	44.7	196.40	0.00	20.64	156.9
T10.01	Potable Water	70.0	92.6	196.40	0.00	17.61	66.2
Total of Fresh Water		300.0		195.55	0.00	18.24	343.4

Location _	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB01.03	Water Ballast 01.03	353.9	100.0	3.19	0.00	11.41	0.0
WB01.11	Water Ballast 01.11	772.1	94.7	7.20	17.83	19.96	152.8
WB01.12	Water Ballast 01.12	747.6	91.7	7.20	-17.83	19.74	152.8
WB02.05	Water Ballast 02.05	1524.8	100.0	23.01	5.35	7.78	0.0
WB02.06	Water Ballast 02.06	1526.6	100.0	23.02	-5.35	7.78	0.0
WB02.07	Water Ballast 02.07	946.7	99.4	24.38	15.77	8.36	0.0
WB02.08	Water Ballast 02.08	946.7	99.4	24.38	-15.77	8.36	0.0
WB02.09	Water Ballast 02.09	657.6	100.0	20.51	5.65	11.90	0.0
WB02.10	Water Ballast 02.10	657.6	100.0	20.51	-5.65	11.90	0.0
WB02.11	Water Ballast 02.11	562.9	100.0	20.84	16.51	11.95	0.0
WB02.12	Water Ballast 02.12	562.9	100.0	20.84	-16.51	11.95	0.0
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	2128.4	100.0	46.50	16.11	6.30	0.0
WB03.08	Water Ballast 03.08	2128.4	100.0	46.50	-16.11	6.30	0.0
WB03.09	Water Ballast 03.09	603.3	100.0	45.60	5.69	11.90	0.0
WB03.10	Water Ballast 03.10	603.3	100.0	45.60	-5.69	11.90	0.0
WB03.11	Water Ballast 03.11	533.4	100.0	45.66	16.41	11.90	0.0
WB03.12	Water Ballast 03.12	533.4	100.0	45.66	-16.41	11.90	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.08	Water Ballast 04.08	2001.6	100.0	69.60	-16.44	6.70	0.0
WB04.09	Water Ballast 04.09	603.3	100.0	69.60	5.69	11.90	0.0
WB04.10	Water Ballast 04.10	603.3	100.0	69.60	-5.69	11.90	0.0
WB04.11	Water Ballast 04.11	537.0	100.0	69.60	16.44	11.90	0.0
WB04.12	Water Ballast 04.12	537.0	100.0	69.60	-16.44	11.90	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.05	Water Ballast 05.05	627.1	29.0	93.41	5.59	3.77	2956.9
WB05.06	Water Ballast 05.06	627.1	29.0	93.41	-5.59	3.77	2956.9
WB05.07	Water Ballast 05.07	2001.6	100.0	93.60	16.44	6.70	0.0
WB05.08	Water Ballast 05.08	2001.6	100.0	93.60	-16.44	6.70	0.0
WB05.09	Water Ballast 05.09	603.3	100.0	93.60	5.69	11.90	0.0
WB05.10	Water Ballast 05.10	603.3	100.0	93.60	-5.69	11.90	0.0
WB05.11	Water Ballast 05.11	532.8	100.0	93.66	16.40	11.90	0.0
WB05.12	Water Ballast 05.12	532.8	100.0	93.66	-16.40	11.90	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.05	Water Ballast 06.05	2158.9	100.0	117.53	5.48	6.71	0.0
WB06.06	Water Ballast 06.06	2158.9	100.0	117.53	-5.48	6.71	0.0
WB06.07	Water Ballast 06.07	2001.6	100.0	117.60	16.44	6.70	0.0
WB06.08	Water Ballast 06.08	2001.6	100.0	117.60	-16.44	6.70	0.0
WB06.09	Water Ballast 06.09	603.3	100.0	117.60	5.69	11.90	0.0
WB06.10	Water Ballast 06.10	603.3	100.0	117.60	-5.69	11.90	0.0
WB06.11	Water Ballast 06.11	537.0	100.0	117.60	16.44	11.90	0.0
WB06.12	Water Ballast 06.12	537.0	100.0	117.60	-16.44	11.90	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB07.05	Water Ballast 07.05	2599.8	100.0	143.95	5.48	6.72	0.0
WB07.06	Water Ballast 07.06	2599.8	100.0	143.95	-5.48	6.72	0.0
WB07.07	Water Ballast 07.07	2390.6	100.0	143.94	16.42	6.71	0.0
WB07.08	Water Ballast 07.08	2390.6	100.0	143.94	-16.42	6.71	0.0
WB07.09	Water Ballast 07.09	715.7	100.0	143.84	5.69	11.91	0.0
WB07.10	Water Ballast 07.10	715.7	100.0	143.84	-5.69	11.91	0.0
WB07.11	Water Ballast 07.11	640.8	100.0	144.06	16.41	11.90	0.0
WB07.12	Water Ballast 07.12	640.8	100.0	144.06	-16.41	11.90	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0

WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	1199.6	100.0	167.13	15.42	7.04	0.0
WB08.08	Water Ballast 08.08	1199.6	100.0	167.13	-15.42	7.04	0.0
WB08.09	Water Ballast 08.09	351.0	100.0	168.00	5.69	12.20	0.0
WB08.10	Water Ballast 08.10	351.0	100.0	168.00	-5.69	12.20	0.0
WB08.11	Water Ballast 08.11	391.3	100.0	167.62	16.01	11.90	0.0
WB08.12	Water Ballast 08.12	391.3	100.0	167.62	-16.01	11.90	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	633.4	100.0	183.26	12.72	8.16	0.0
WB09.08	Water Ballast 09.08	633.4	100.0	183.26	-12.72	8.16	0.0
WB09.09	Water Ballast 09.09	330.8	100.0	183.21	0.00	12.20	0.0
WB09.11	Water Ballast 09.11	1466.5	100.0	182.74	14.97	20.53	0.0
WB09.12	Water Ballast 09.12	1466.4	100.0	182.74	-14.97	20.53	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB10.05	Water Ballast 10.05	1046.0	100.0	194.21	5.79	8.04	0.0
WB10.06	Water Ballast 10.06	1046.0	100.0	194.25	-5.75	8.04	0.0
WB10.11	Water Ballast 10.11	956.2	100.0	193.86	11.70	20.71	0.0
WB10.12	Water Ballast 10.12	956.2	100.0	193.86	-11.70	20.71	0.0
WB11.01	Water Ballast 11.01	1639.7	100.0	205.41	-0.01	7.51	0.0
WB11.11	Water Ballast 11.11	1713.7	90.1	205.53	5.01	21.78	2654.1
WB11.12	Water Ballast 11.12	1645.1	84.0	205.23	-4.71	21.21	2654.4
Total of Ballast Water		84284.0		113.41	0.01	8.54	11527.8

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Stores							
(STO)		100.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
Deadweight		88705.1		111.89	0.00	8.48	14464.8
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		109576.5		112.18	-0.00	8.85	14464.8

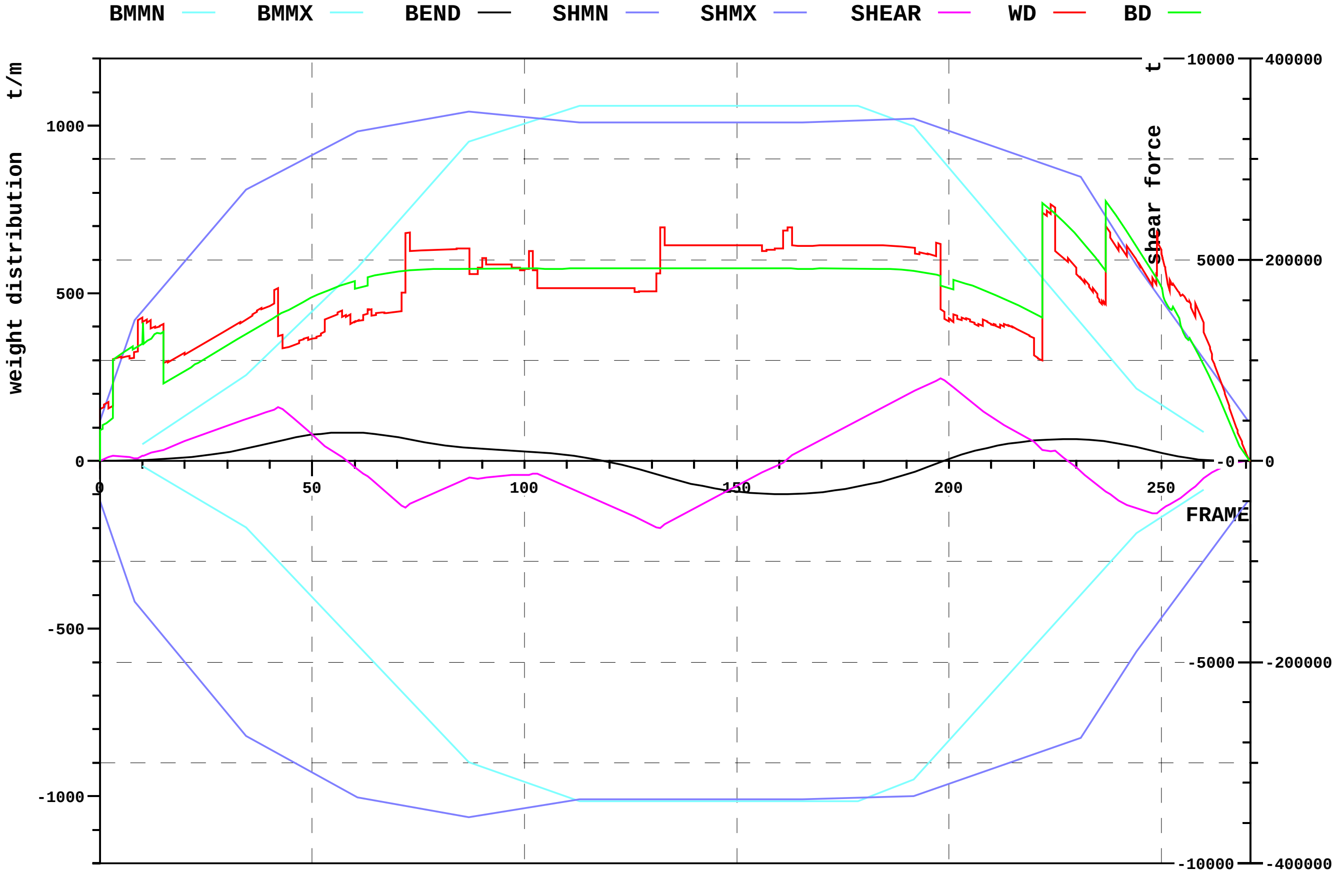


RCR	TEXT	REQ	ATTV	UNIT	STAT
MINGM0.3-SUB	GM>0.3m	0.300	0.774	m	OK
RANGE-SUB	Range of Stability	7.000	60.000	deg	OK
MAXGZ-SUB	Max GZ	0.050	0.203	m	OK
MINGM0.5-SUB	GM>0.5m	0.500	0.774	m	OK

Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	7.350
AIR_DUCT_1	UNPROTECTED	28.8	222.00	11.375	33.350	7.350
AIR_DUCT_3	UNPROTECTED	52.7	230.00	0.000	33.350	7.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

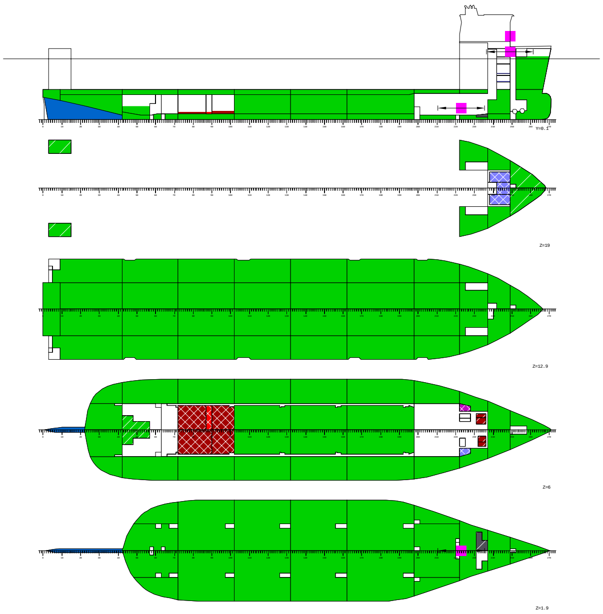
			x	frame
Shear force (min)	-1665.5 t	position:	105.6 m	132
Shear force (max)	2056.6 t		158.4 m	198
Max. rel. shear force	33.0 %		199.2 m	249
Sagging moment	-33003.0 tm		129.0 m	161
Hogging moment	28233.1 tm		46.4 m	58
Max. rel. bending moment	47.4 %		8.0 m	10

LOADING CONDITION LC41, Submerging condition 26m, departure

FR	X	BEND	BMMN	BMMX	RELBM
#	m	tm	tm	tm	%
21.00	16.80	3740	-32571	47578	9.4
42.00	33.60	19537	-100351	116458	10.6
57.00	45.60	28173	-167096	177641	13.3
72.00	57.60	22023	-233865	246782	6.5
87.00	69.60	12987	-300085	317780	1.3
102.00	81.60	8561	-322267	337935	0.2
117.00	93.60	1283	-338607	352783	1.7
132.00	105.60	-14394	-338607	352783	6.2
147.00	117.60	-28673	-338607	352783	10.3
162.00	129.60	-32990	-338607	352783	11.6
180.00	144.00	-24357	-336305	350692	9.2
198.00	158.40	-1401	-287350	301648	2.9
210.00	168.00	13982	-231392	242029	3.7
222.00	177.60	20963	-175443	182421	9.8
237.00	189.60	19293	-105528	107934	16.9
249.00	199.20	8899	-58776	58942	15.0

FR	X	SHEAR	SHMN	SHMX	RELSH
#	m	t	t	t	%
21.00	16.80	530	-5135	5093	10.8
42.00	33.60	1341	-7273	7164	19.3
57.00	45.60	98	-8146	7989	2.2
72.00	57.60	-1162	-8573	8405	12.7
87.00	69.60	-409	-8852	8685	3.7
102.00	81.60	-315	-8599	8528	3.3
117.00	93.60	-965	-8412	8412	11.5
132.00	105.60	-1665	-8412	8412	19.8
147.00	117.60	-782	-8412	8412	9.3
162.00	129.60	42	-8412	8412	0.5
180.00	144.00	1089	-8370	8470	12.3
198.00	158.40	2057	-8104	8286	24.0
210.00	168.00	1104	-7660	7841	13.1
222.00	177.60	277	-7215	7397	2.6
237.00	189.60	-768	-5918	6074	14.1
249.00	199.20	-1306	-4059	4161	33.0

2.3 LC42 Submerging condition 26m, arrival



LOADING CONDITION LC42: Submerging condition 26m, arrival

FLOATING POSITION

Mean draught (moulded)	26.00 m	KM above the moulded base	9.76 m
Draught at AP (moulded)	26.00 m	KG above the moulded base	8.78 m
Draught at FP (moulded)	26.00 m	GM0 (solid)	0.98 m
Trim	0.00 m	Free surface correction	-0.12 m
Heeling	0.0 deg		
		GM (fluid)	0.87 m
Mean draught (heel 0)	26.00 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5445.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	87440.6	112.11	-0.00	8.44	6628.3
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	88704.8	111.89	0.00	8.39	12710.0
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	109576.2	112.18	-0.00	8.78	12710.0

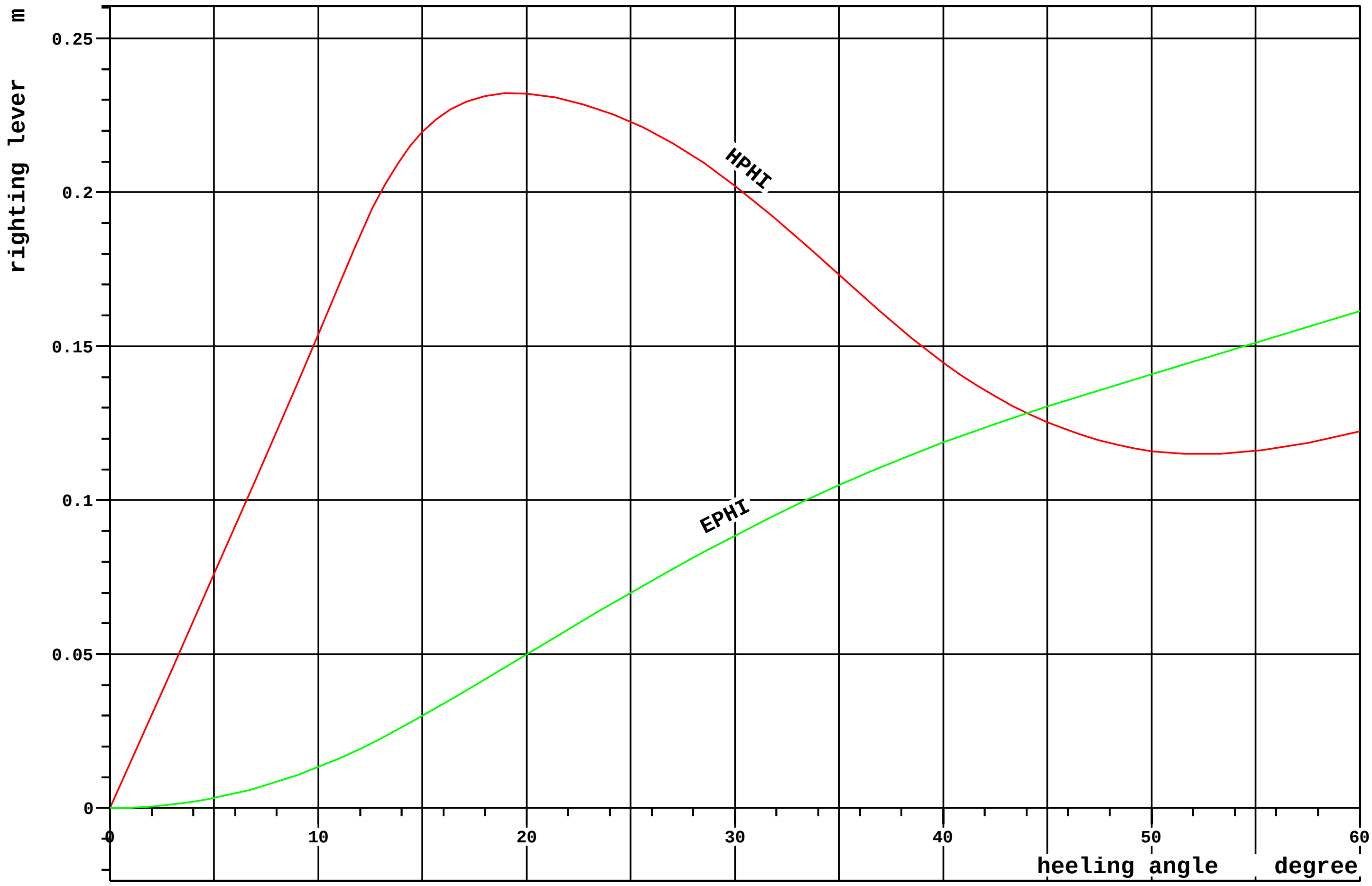
LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HFO1	HFO Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HFO2	HFO Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HFO3	HFO Storage Tank 3	99.2	12.0	76.61	5.36	3.10	1143.5
HFO4	HFO Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	1143.5
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5445.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
_							
Ballast Water density=1.025 t/m3							
WB01.03	Water Ballast 01.03	353.9	100.0	3.19	0.00	11.41	0.0
WB01.11	Water Ballast 01.11	758.0	93.0	7.20	17.83	19.84	152.8
WB01.12	Water Ballast 01.12	758.0	93.0	7.20	-17.83	19.84	152.8
WB02.05	Water Ballast 02.05	1524.8	100.0	23.01	5.35	7.78	0.0
WB02.06	Water Ballast 02.06	1526.6	100.0	23.02	-5.35	7.78	0.0
WB02.07	Water Ballast 02.07	946.7	99.4	24.38	15.77	8.36	0.0
WB02.08	Water Ballast 02.08	946.7	99.4	24.38	-15.77	8.36	0.0
WB02.09	Water Ballast 02.09	657.6	100.0	20.51	5.65	11.90	0.0
WB02.10	Water Ballast 02.10	657.6	100.0	20.51	-5.65	11.90	0.0
WB02.11	Water Ballast 02.11	562.9	100.0	20.84	16.51	11.95	0.0
WB02.12	Water Ballast 02.12	562.9	100.0	20.84	-16.51	11.95	0.0
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.06	Water Ballast 03.06	370.1	38.5	39.19	0.00	4.28	1014.3
WB03.07	Water Ballast 03.07	2128.4	100.0	46.50	16.11	6.30	0.0
WB03.08	Water Ballast 03.08	2128.4	100.0	46.50	-16.11	6.30	0.0
WB03.09	Water Ballast 03.09	603.3	100.0	45.60	5.69	11.90	0.0
WB03.10	Water Ballast 03.10	603.3	100.0	45.60	-5.69	11.90	0.0
WB03.11	Water Ballast 03.11	533.4	100.0	45.66	16.41	11.90	0.0
WB03.12	Water Ballast 03.12	533.4	100.0	45.66	-16.41	11.90	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.08	Water Ballast 04.08	2001.6	100.0	69.60	-16.44	6.70	0.0
WB04.09	Water Ballast 04.09	603.3	100.0	69.60	5.69	11.90	0.0
WB04.10	Water Ballast 04.10	603.3	100.0	69.60	-5.69	11.90	0.0
WB04.11	Water Ballast 04.11	537.0	100.0	69.60	16.44	11.90	0.0
WB04.12	Water Ballast 04.12	537.0	100.0	69.60	-16.44	11.90	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.05	Water Ballast 05.05	2158.9	100.0	93.53	5.48	6.71	0.0
WB05.06	Water Ballast 05.06	2158.9	100.0	93.53	-5.48	6.71	0.0
WB05.07	Water Ballast 05.07	2001.6	100.0	93.60	16.44	6.70	0.0
WB05.08	Water Ballast 05.08	2001.6	100.0	93.60	-16.44	6.70	0.0
WB05.09	Water Ballast 05.09	603.3	100.0	93.60	5.69	11.90	0.0
WB05.10	Water Ballast 05.10	603.3	100.0	93.60	-5.69	11.90	0.0
WB05.11	Water Ballast 05.11	532.8	100.0	93.66	16.40	11.90	0.0
WB05.12	Water Ballast 05.12	532.8	100.0	93.66	-16.40	11.90	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.05	Water Ballast 06.05	2158.9	100.0	117.53	5.48	6.71	0.0
WB06.06	Water Ballast 06.06	2158.9	100.0	117.53	-5.48	6.71	0.0
WB06.07	Water Ballast 06.07	2001.6	100.0	117.60	16.44	6.70	0.0
WB06.08	Water Ballast 06.08	2001.6	100.0	117.60	-16.44	6.70	0.0
WB06.09	Water Ballast 06.09	603.3	100.0	117.60	5.69	11.90	0.0
WB06.10	Water Ballast 06.10	603.3	100.0	117.60	-5.69	11.90	0.0
WB06.11	Water Ballast 06.11	537.0	100.0	117.60	16.44	11.90	0.0
WB06.12	Water Ballast 06.12	537.0	100.0	117.60	-16.44	11.90	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB07.05	Water Ballast 07.05	2599.8	100.0	143.95	5.48	6.72	0.0
WB07.06	Water Ballast 07.06	2599.8	100.0	143.95	-5.48	6.72	0.0
WB07.07	Water Ballast 07.07	2390.6	100.0	143.94	16.42	6.71	0.0
WB07.08	Water Ballast 07.08	2390.6	100.0	143.94	-16.42	6.71	0.0
WB07.09	Water Ballast 07.09	715.7	100.0	143.84	5.69	11.91	0.0
WB07.10	Water Ballast 07.10	715.7	100.0	143.84	-5.69	11.91	0.0
WB07.11	Water Ballast 07.11	640.8	100.0	144.06	16.41	11.90	0.0
WB07.12	Water Ballast 07.12	640.8	100.0	144.06	-16.41	11.90	0.0

WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	1199.6	100.0	167.13	15.42	7.04	0.0
WB08.08	Water Ballast 08.08	1199.6	100.0	167.13	-15.42	7.04	0.0
WB08.09	Water Ballast 08.09	351.0	100.0	168.00	5.69	12.20	0.0
WB08.10	Water Ballast 08.10	351.0	100.0	168.00	-5.69	12.20	0.0
WB08.11	Water Ballast 08.11	391.3	100.0	167.62	16.01	11.90	0.0
WB08.12	Water Ballast 08.12	391.3	100.0	167.62	-16.01	11.90	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	633.4	100.0	183.26	12.72	8.16	0.0
WB09.08	Water Ballast 09.08	633.4	100.0	183.26	-12.72	8.16	0.0
WB09.09	Water Ballast 09.09	330.8	100.0	183.21	0.00	12.20	0.0
WB09.11	Water Ballast 09.11	1466.5	100.0	182.74	14.97	20.53	0.0
WB09.12	Water Ballast 09.12	1466.4	100.0	182.74	-14.97	20.53	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB10.05	Water Ballast 10.05	1046.0	100.0	194.21	5.79	8.04	0.0
WB10.06	Water Ballast 10.06	1046.0	100.0	194.25	-5.75	8.04	0.0
WB10.11	Water Ballast 10.11	956.2	100.0	193.86	11.70	20.71	0.0
WB10.12	Water Ballast 10.12	956.2	100.0	193.86	-11.70	20.71	0.0
WB11.01	Water Ballast 11.01	1639.7	100.0	205.41	-0.01	7.51	0.0
WB11.11	Water Ballast 11.11	1500.7	78.9	205.30	4.74	20.89	2654.1
WB11.12	Water Ballast 11.12	1584.7	80.9	205.16	-4.63	20.96	2654.4
Total of Ballast Water		87440.6		112.11	-0.00	8.44	6628.3

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Stores							
(STO)		10.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
Deadweight		88704.8		111.89	0.00	8.39	12710.0
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		109576.2		112.18	-0.00	8.78	12710.0

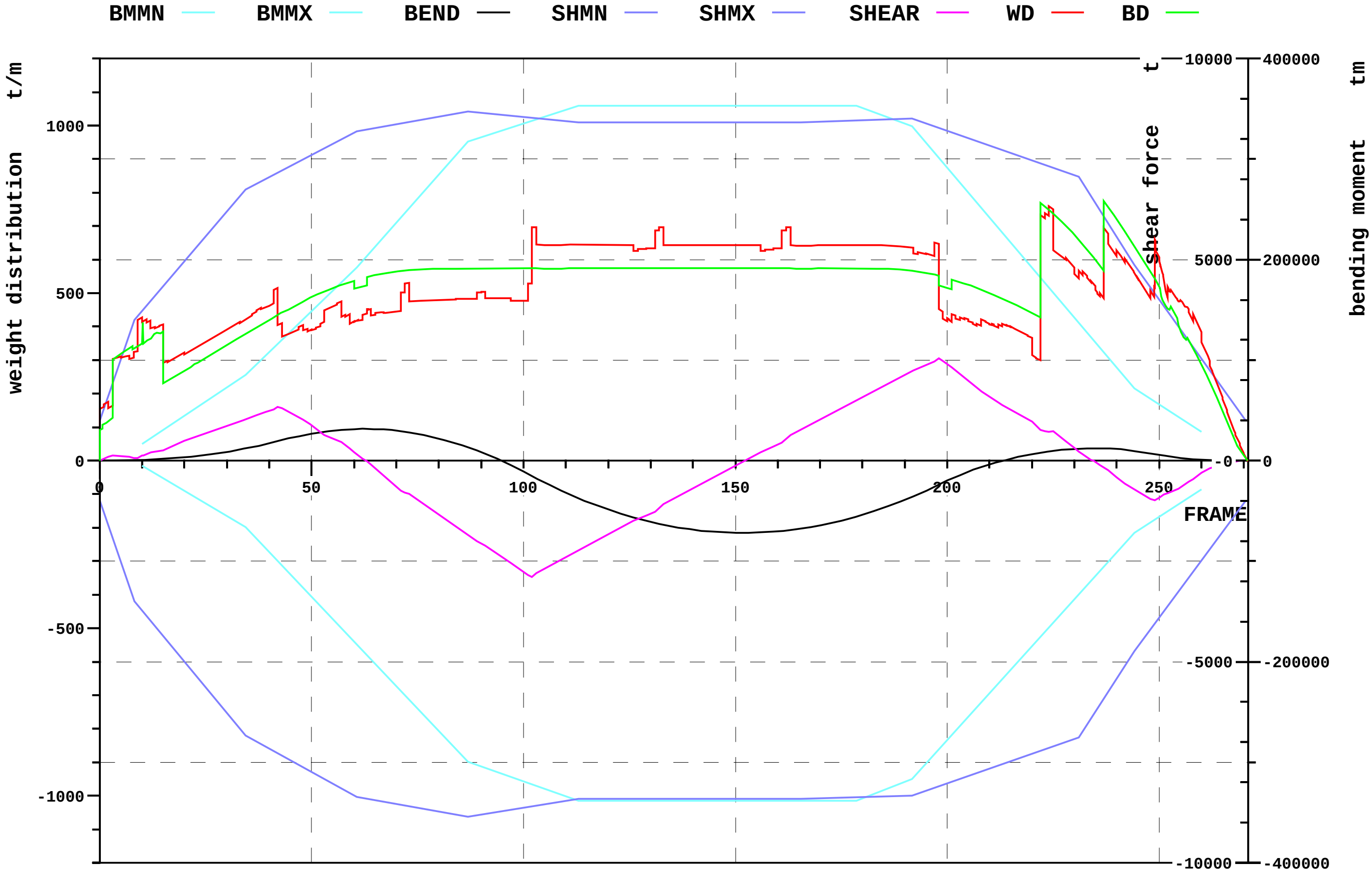


RCR	TEXT	REQ	ATTV	UNIT	STAT
MINGM0.3-SUB	GM>0.3m	0.300	0.866	m	OK
RANGE-SUB	Range of Stability	7.000	60.000	deg	OK
MAXGZ-SUB	Max GZ	0.050	0.232	m	OK
MINGM0.5-SUB	GM>0.5m	0.500	0.866	m	OK

Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	7.350
AIR_DUCT_1	UNPROTECTED	28.8	222.00	11.375	33.350	7.350
AIR_DUCT_3	UNPROTECTED	52.7	230.00	0.000	33.350	7.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements   weight: 20871.4 t   LCG: 113.44 m

			x	frame
Shear force (min)	-2887.3 t	position:	81.6 m	102
Shear force (max)	2546.8 t		158.4 m	198
Max. rel. shear force	33.3 %		81.6 m	102
Sagging moment	-71776.7 tm		122.4 m	153
Hogging moment	31585.4 tm		50.4 m	63
Max. rel. bending moment	47.4 %		8.0 m	10

LOADING CONDITION LC42, Submerging condition 26m, arrival

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELBM %
21.00	16.80	3724	-32571	47578	9.4
42.00	33.60	19477	-100351	116458	10.5
57.00	45.60	30452	-167096	177641	14.6
72.00	57.60	28708	-233865	246782	9.3
87.00	69.60	13220	-300085	317780	1.4
102.00	81.60	-15055	-322267	337935	6.9
117.00	93.60	-43968	-338607	352783	14.8
132.00	105.60	-62882	-338607	352783	20.2
147.00	117.60	-71257	-338607	352783	22.7
162.00	129.60	-69671	-338607	352783	22.2
180.00	144.00	-53958	-336305	350692	17.8
198.00	158.40	-23924	-287350	301648	10.6
210.00	168.00	-3825	-231392	242029	3.9
222.00	177.60	7873	-175443	182421	2.4
237.00	189.60	12260	-105528	107934	10.4
249.00	199.20	6514	-58776	58942	10.9

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	526	-5135	5093	10.7
42.00	33.60	1337	-7273	7164	19.3
57.00	45.60	465	-8146	7989	6.7
72.00	57.60	-796	-8573	8405	8.4
87.00	69.60	-1853	-8852	8685	20.2
102.00	81.60	-2887	-8599	8528	33.3
117.00	93.60	-2001	-8412	8412	23.8
132.00	105.60	-1175	-8412	8412	14.0
147.00	117.60	-291	-8412	8412	3.5
162.00	129.60	532	-8412	8412	6.3
180.00	144.00	1580	-8370	8470	18.2
198.00	158.40	2547	-8104	8286	30.0
210.00	168.00	1594	-7660	7841	19.4
222.00	177.60	767	-7215	7397	9.3
237.00	189.60	-176	-5918	6074	4.2
249.00	199.20	-990	-4059	4161	25.3

3 Aftload and sideload conditions

3.1 Summary of loading conditions

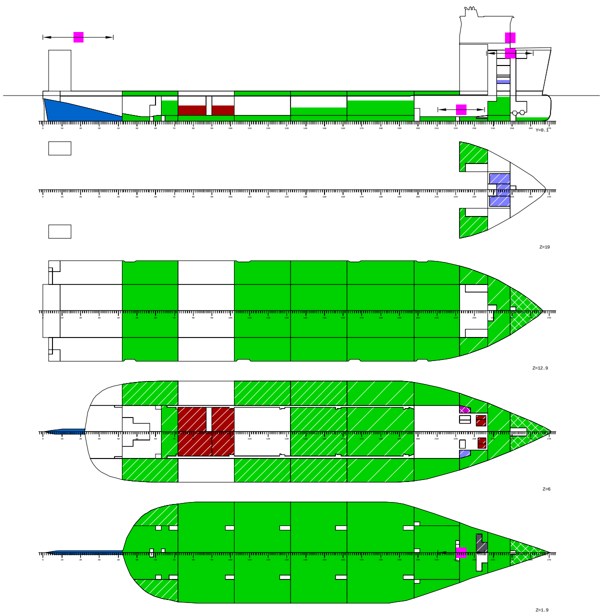
DELTAMARIN CONTRACTING LTD
Purokatu 1
FIN-21200 RAISIO FINLAND
www.deltamarin.com
SFS-EN ISO 9001:2000 No 1437-07
Business ID: 0595243-0
VAT number: FI05952430
Domicile: Raisio
Tel. +358-2-4336 300
Fax +358-2-4380 378
E-mail: info@deltamarin.com

CASE	TEXT	DISP t	DWT t	BW t	CARGO t	T m	TR m	HEEL degree	KMT m
AFTLOAD1	Non-Buoyant Cargo, half loaded from aft	80381.5	59510.1	46521.3	10000.0	11.000	0.000	0.0	21.123
AFTLOAD2	Non-Buoyant Cargo, aft load	80380.5	59509.1	36520.2	20000.0	11.000	0.000	0.0	21.123
SIDELOAD1	Non-Buoyant Cargo, half loaded from side	80382.4	59511.0	46522.2	10000.0	11.000	0.000	0.0	21.123

CASE	KGF m	GM m	GMCORR m	BMMIN tm	BMMAX tm	RELBM %	XRELBM m	SHMIN t	SHMAX t	RELSH %	XRELSH m
AFTLOAD1	11.54	9.581	0.542	-2770	314167	74.2	39	-4490	7364	97.5	20
AFTLOAD2	15.66	5.466	0.822	-0	472950	100.0	78	-7780	8125	97.5	20
SIDELOAD1	10.84	10.285	0.434	-128317	16526	55.3	8	-3807	2925	48.0	75

CASE	name	
TEXT	descriptive name	
DISP	total displacement	t
DWT	mass of load	t
BW	mass of load	t
CARGO	mass of load	t
T	draught, moulded	m
TR	trim	m
HEEL	heeling angle	degree
KMT	transv. metac. height	m
KGF	cgz of load	m
GM	metacentric height	m
GMCORR	GM correction	m
BMMIN	minimum bending moment	tm
BMMAX	maximum bending moment	tm
RELBM	relative bending moment	%
XRELBM	x where max. relative bending mom.	m
SHMIN	minimum shear force	t
SHMAX	maximum shear force	t
RELSH	relative shear force	%
XRELSH	x where max. relative shear force	m

3.2 AFTLOAD1



LOADING CONDITION AFTLOAD1: Non-Buoyant Cargo, half loaded from aft

FLOATING POSITION

Mean draught (moulded)	11.00 m	KM above the moulded base	21.12 m
Draught at AP (moulded)	11.00 m	KG above the moulded base	11.00 m
Draught at FP (moulded)	11.00 m	GM0 (solid)	10.12 m
Trim	0.00 m	Free surface correction	-0.54 m
Heeling	0.0 deg		
		GM (fluid)	9.58 m
Mean draught (heel 0)	11.00 m		
Trim (heel 0)	0.00 m		

LOADS

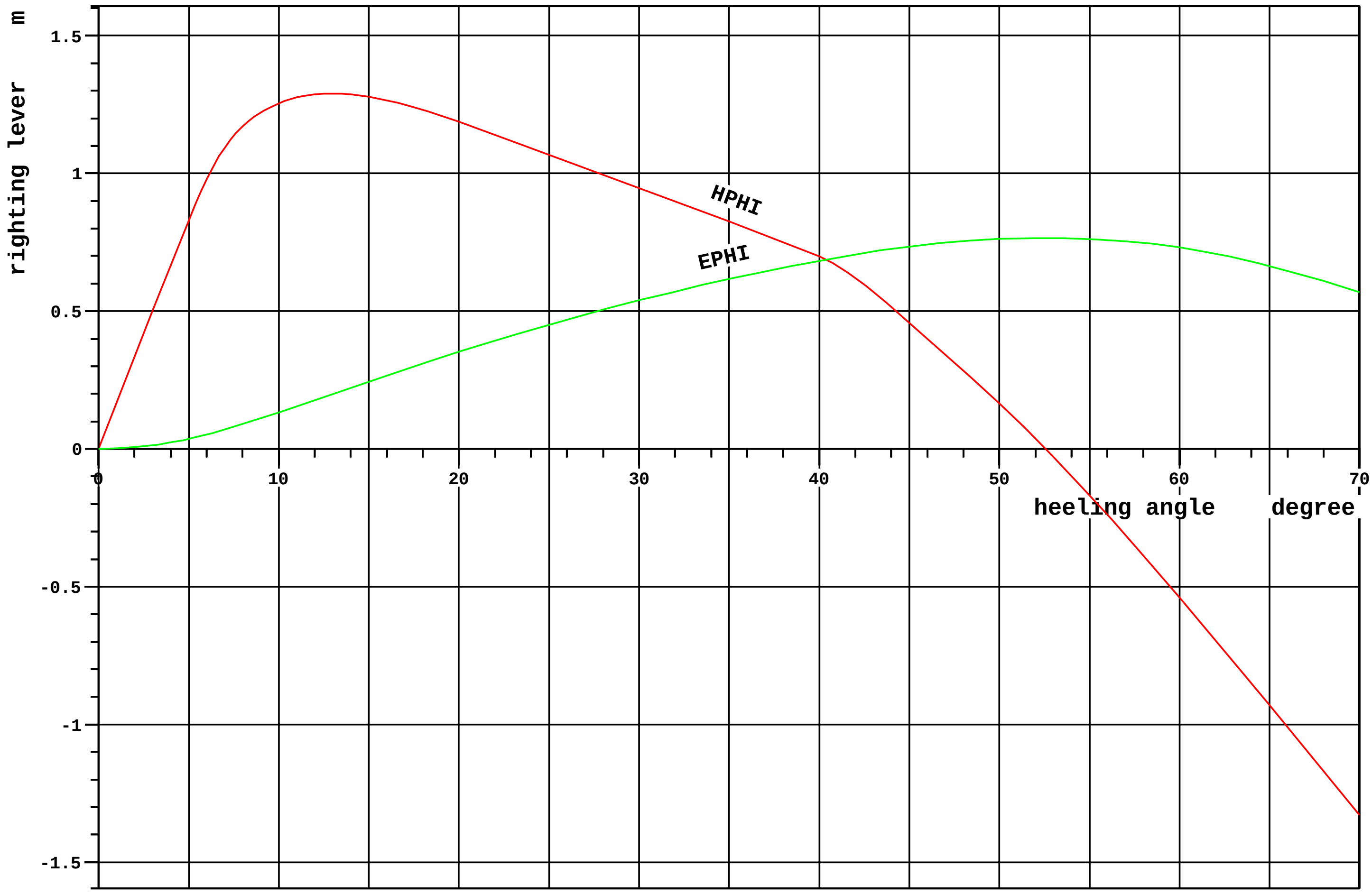
Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	2067.1	74.16	-0.26	4.58	4069.2
Diesel Oil	42.2	179.80	0.00	2.88	14.0
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	57.9	190.64	-3.24	13.31	76.3
Fresh Water	219.2	194.82	0.00	17.92	74.6
Ballast Water	46521.3	124.01	0.01	6.17	38896.0
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	24.7	180.00	5.25	6.25	5.0
Bilge Water	20.4	187.24	-2.65	1.91	133.0
Sludge	36.2	186.86	3.31	1.46	315.1
Solid cargo	10000.0	15.00	0.00	36.00	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	59510.1	104.04	0.00	11.19	43592.4
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	80381.5	106.48	-0.00	11.00	43592.4

LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HFO1	HFO Storage Tank 1	526.0	50.0	63.60	5.55	4.60	1084.6
HFO2	HFO Storage Tank 2	631.2	50.0	64.80	-5.55	4.60	1301.5
HFO3	HFO Storage Tank 3	413.4	50.0	76.71	5.40	4.65	834.7
HFO4	HFO Storage Tank 4	413.4	50.0	76.71	-5.40	4.65	834.7
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		2067.1		74.16	-0.26	4.58	4069.2
Diesel Oil density=0.900 t/m3							
T09.07	MDO Storage P	15.5	50.0	179.30	9.58	2.88	6.4
T09.08	MDO Storage S	15.5	50.0	179.30	-9.58	2.88	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.87	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.87	0.6
Total of Diesel Oil		42.2		179.80	0.00	2.88	14.0
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.1	50.0	179.80	-9.33	6.25	10.1
T10.01	Potable Water	37.8	50.0	196.40	0.00	17.06	66.2
Total of Technical Water		57.9		190.64	-3.24	13.31	76.3
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	109.6	50.0	194.82	4.84	17.92	37.3
FW2	Fresh Water S	109.6	50.0	194.82	-4.84	17.92	37.3
Total of Fresh Water		219.2		194.82	0.00	17.92	74.6

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.05	Water Ballast 03.05	969.5	76.4	53.94	0.00	5.73	5543.8
WB03.07	Water Ballast 03.07	532.1	25.0	48.78	15.37	2.64	1590.7
WB03.08	Water Ballast 03.08	532.1	25.0	48.78	-15.37	2.64	1590.7
WB03.09	Water Ballast 03.09	603.3	100.0	45.60	5.69	11.90	0.0
WB03.10	Water Ballast 03.10	603.3	100.0	45.60	-5.69	11.90	0.0
WB03.11	Water Ballast 03.11	533.4	100.0	45.66	16.41	11.90	0.0
WB03.12	Water Ballast 03.12	533.4	100.0	45.66	-16.41	11.90	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.07	Water Ballast 05.07	1000.8	50.0	93.60	16.44	4.65	2085.3
WB05.08	Water Ballast 05.08	1000.8	50.0	93.60	-16.44	4.65	2085.3
WB05.09	Water Ballast 05.09	603.3	100.0	93.60	5.69	11.90	0.0
WB05.10	Water Ballast 05.10	603.3	100.0	93.60	-5.69	11.90	0.0
WB05.11	Water Ballast 05.11	532.8	100.0	93.66	16.40	11.90	0.0
WB05.12	Water Ballast 05.12	532.8	100.0	93.66	-16.40	11.90	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.05	Water Ballast 06.05	863.6	40.0	117.45	5.56	4.21	2209.8
WB06.06	Water Ballast 06.06	863.6	40.0	117.45	-5.56	4.21	2209.8
WB06.07	Water Ballast 06.07	800.6	40.0	117.60	16.44	4.24	2085.3
WB06.08	Water Ballast 06.08	800.6	40.0	117.60	-16.44	4.24	2085.3
WB06.09	Water Ballast 06.09	603.3	100.0	117.60	5.69	11.90	0.0
WB06.10	Water Ballast 06.10	603.3	100.0	117.60	-5.69	11.90	0.0
WB06.11	Water Ballast 06.11	537.0	100.0	117.60	16.44	11.90	0.0
WB06.12	Water Ballast 06.12	537.0	100.0	117.60	-16.44	11.90	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB07.05	Water Ballast 07.05	1949.8	75.0	143.88	5.41	5.68	3548.3
WB07.06	Water Ballast 07.06	1949.8	75.0	143.88	-5.41	5.68	3548.3
WB07.07	Water Ballast 07.07	1785.5	74.7	143.92	16.41	5.68	2499.4
WB07.08	Water Ballast 07.08	1785.5	74.7	143.92	-16.41	5.68	2499.4
WB07.09	Water Ballast 07.09	715.7	100.0	143.84	5.69	11.91	0.0
WB07.10	Water Ballast 07.10	715.7	100.0	143.84	-5.69	11.91	0.0
WB07.11	Water Ballast 07.11	640.8	100.0	144.06	16.41	11.90	0.0
WB07.12	Water Ballast 07.12	640.8	100.0	144.06	-16.41	11.90	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	1199.6	100.0	167.13	15.42	7.04	0.0
WB08.08	Water Ballast 08.08	1199.6	100.0	167.13	-15.42	7.04	0.0
WB08.09	Water Ballast 08.09	351.0	100.0	168.00	5.69	12.20	0.0
WB08.10	Water Ballast 08.10	351.0	100.0	168.00	-5.69	12.20	0.0
WB08.11	Water Ballast 08.11	391.3	100.0	167.62	16.01	11.90	0.0
WB08.12	Water Ballast 08.12	391.3	100.0	167.62	-16.01	11.90	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	475.0	75.0	183.40	12.18	6.92	188.9
WB09.08	Water Ballast 09.08	475.0	75.0	183.40	-12.18	6.92	188.9
WB09.11	Water Ballast 09.11	843.2	57.5	182.70	14.65	17.59	924.7
WB09.12	Water Ballast 09.12	806.5	55.0	182.70	-14.63	17.41	905.8
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB10.05	Water Ballast 10.05	784.5	75.0	194.17	5.46	6.79	1249.5
WB10.06	Water Ballast 10.06	784.5	75.0	194.19	-5.43	6.79	1327.1
WB11.01	Water Ballast 11.01	126.3	7.7	204.88	-0.00	1.13	529.6
Total of Ballast Water		46521.3		124.01	0.01	6.17	38896.9

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	9.3	50.0	180.00	4.16	6.25	0.9
T09.13	Gray Water	15.4	50.0	180.00	5.91	6.25	4.1
Total of Gray Water		24.7		180.00	5.25	6.25	5.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	5.2	50.0	188.36	-0.73	3.21	0.6
T09.22	Bilge Water	15.2	50.0	186.86	-3.31	1.46	132.4
Total of Bilge Water		20.4		187.24	-2.65	1.91	133.0
Sludge density=2.380 t/m3							
T09.21	Sludge	36.2	50.0	186.86	3.31	1.46	315.1
Solid cargo density=1.000 t/m3							
(CAS)	NON-BUOYANT CARGO	10000.0		15.00	0.00	36.00	0.0
Stores							
(STO)		100.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
Deadweight		59510.1		104.04	0.00	11.19	43592.4
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		80381.5		106.48	-0.00	11.00	43592.4

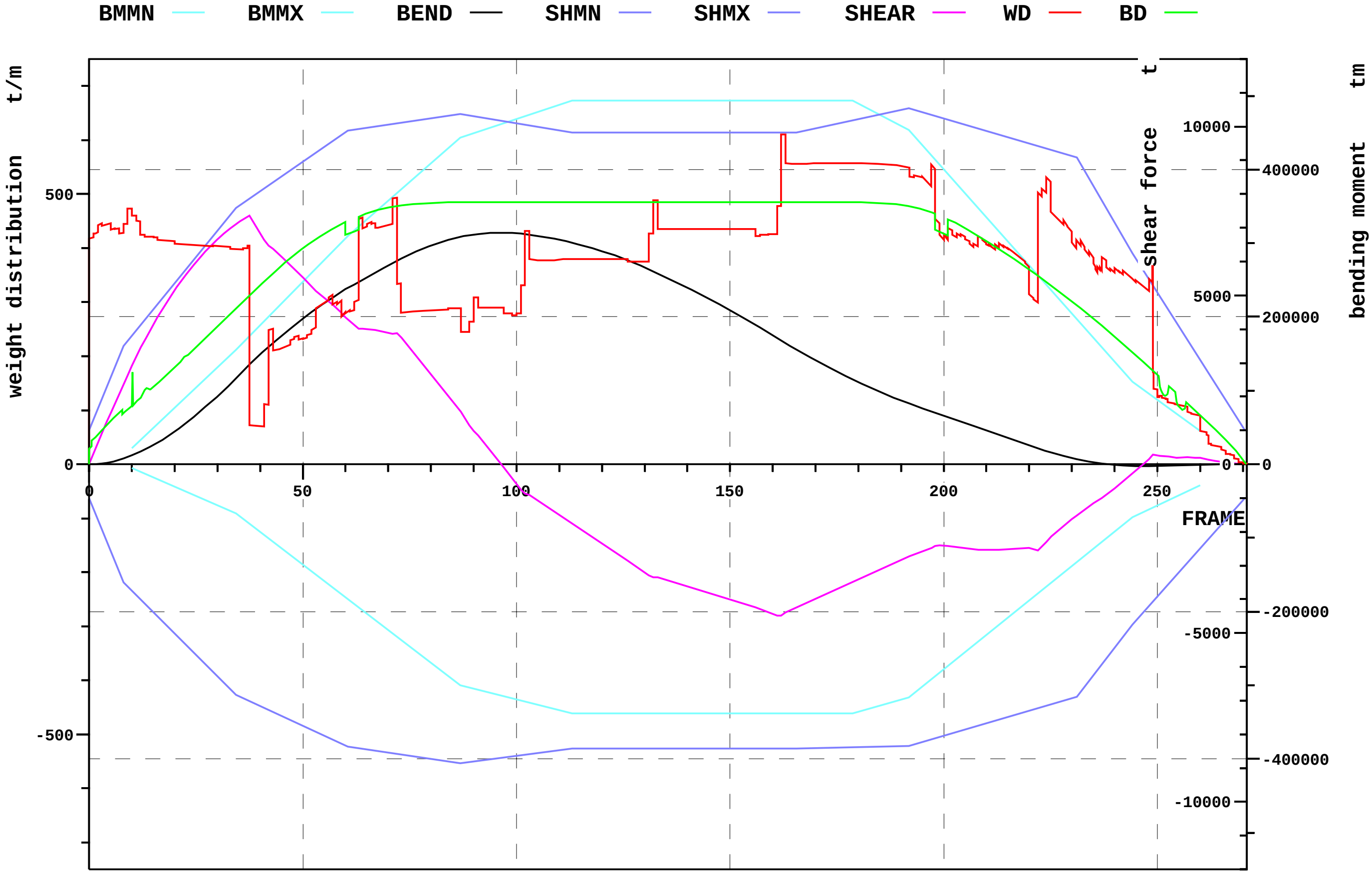


RCR	TEXT	REQ	ATTV	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.072	0.199	mrad	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.144	mrad	OK
V.GZ0.2	Min. GZ > 0.2	0.200	0.947	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	12.961	deg	NOT.
V.GM0.15	GM > 0.15 m	0.150	9.581	m	OK
IMOWEATHER	IMO weather criterion	1.000	2.051		OK
ND-RANGE	Range of Stability acc. ND	36.000	52.573	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	18.381		OK

Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	22.350
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	22.350
AIR_DUCT_1	UNPROTECTED	56.9	222.00	11.375	33.350	22.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

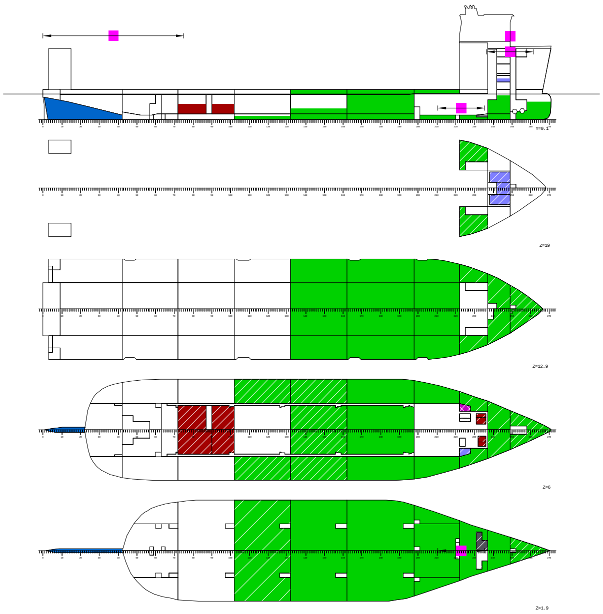
			x	frame
Shear force (min)	-4490.1 t	position:	129.6 m	162
Shear force (max)	7364.2 t		30.0 m	38
Max. rel. shear force	97.5 %		19.6 m	24
Sagging moment	-2770.0 tm		197.3 m	247
Hogging moment	314166.8 tm		77.6 m	97
Max. rel. bending moment	74.2 %		38.9 m	49

LOADING CONDITION AFTLOAD1, Non-Buoyant Cargo, half loaded from aft

FR	X	BEND	BMMN	BMMX	RELBM
#	m	tm	tm	tm	%
21.00	16.80	48852	-32571	82267	41.8
42.00	33.60	159451	-100351	200325	72.8
57.00	45.60	226481	-167096	288693	72.7
72.00	57.60	275843	-233865	367863	69.4
87.00	69.60	308596	-300085	443551	63.7
102.00	81.60	312172	-322267	472486	59.7
117.00	93.60	294537	-338607	493802	52.1
132.00	105.60	261693	-338607	493802	44.2
147.00	117.60	218493	-338607	493802	33.9
162.00	129.60	168130	-338607	493802	21.8
180.00	144.00	111487	-336305	489619	8.4
198.00	158.40	69674	-287350	412939	2.0
210.00	168.00	45964	-231392	334884	2.0
222.00	177.60	21782	-175443	256842	8.8
237.00	189.60	1234	-105528	159320	19.4
249.00	199.20	-2518	-58776	91901	25.3

FR	X	SHEAR	SHMN	SHMX	RELSH
#	m	t	t	t	%
21.00	16.80	5332	-5135	5507	96.7
42.00	33.60	6467	-7273	8253	77.0
57.00	45.60	4719	-8146	9561	45.3
72.00	57.60	3876	-8573	10092	33.4
87.00	69.60	1548	-8852	10370	8.2
102.00	81.60	-872	-8599	10055	17.2
117.00	93.60	-2100	-8412	9823	30.8
132.00	105.60	-3346	-8412	9823	44.4
147.00	117.60	-3891	-8412	9823	50.4
162.00	129.60	-4490	-8412	9823	57.0
180.00	144.00	-3411	-8370	10215	46.6
198.00	158.40	-2422	-8104	10302	38.3
210.00	168.00	-2533	-7660	9858	41.5
222.00	177.60	-2560	-7215	9413	44.0
237.00	189.60	-1003	-5918	7808	28.4
249.00	199.20	286	-4059	5297	7.1

3.3 AFTLOAD2



LOADING CONDITION AFTLOAD2: Non-Buoyant Cargo, aft load

FLOATING POSITION

Mean draught (moulded)	11.00 m	KM above the moulded base	21.12 m
Draught at AP (moulded)	11.00 m	KG above the moulded base	14.83 m
Draught at FP (moulded)	11.00 m	GM0 (solid)	6.29 m
Trim	0.00 m	Free surface correction	-0.82 m
Heeling	0.0 deg		
		GM (fluid)	5.47 m
Mean draught (heel 0)	11.00 m		
Trim (heel 0)	0.00 m		

LOADS

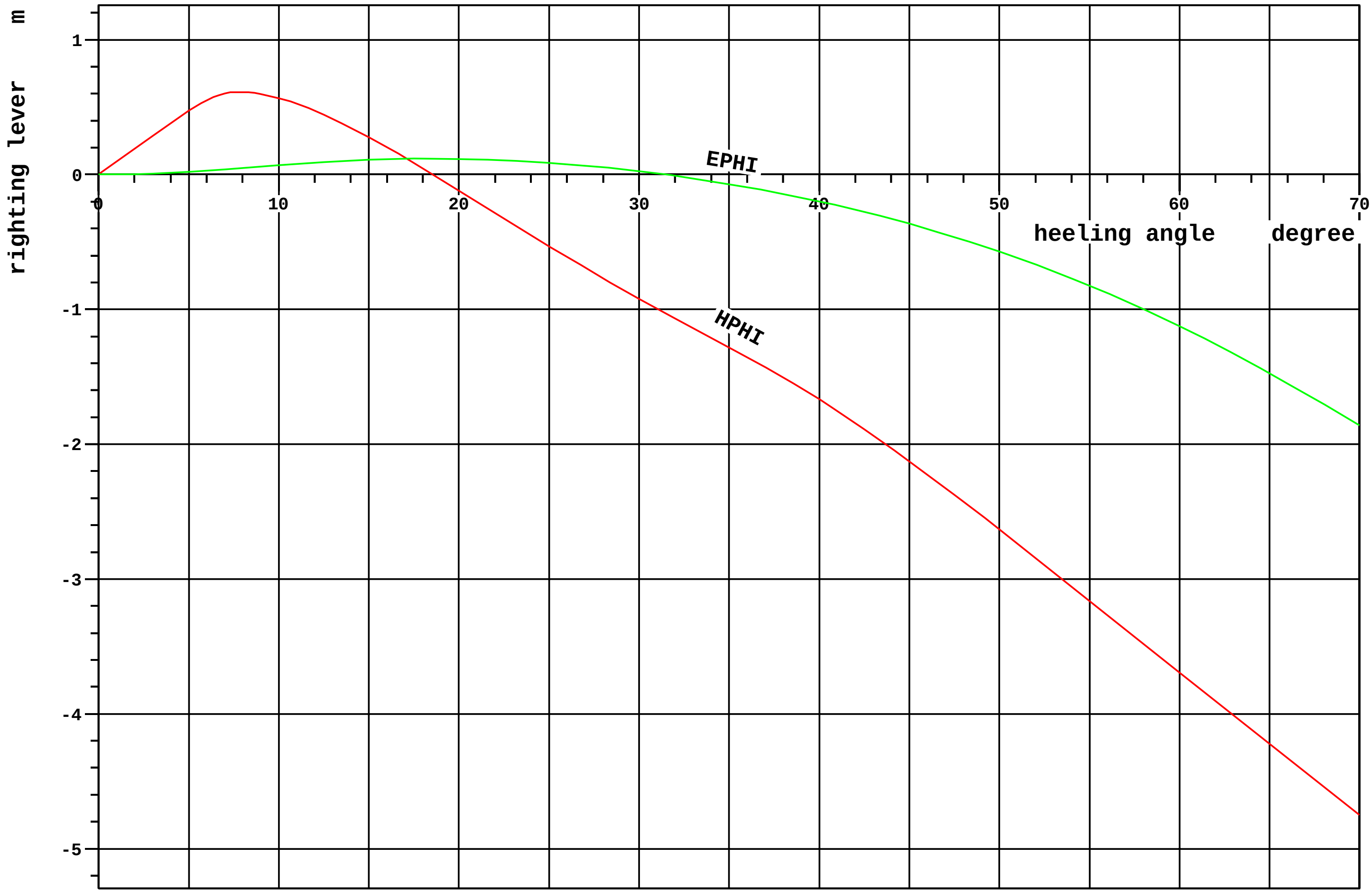
Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	2067.1	74.16	-0.26	4.58	4069.2
Diesel Oil	42.2	179.80	0.00	2.88	14.0
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	57.9	190.64	-3.24	13.31	76.3
Fresh Water	219.2	194.82	0.00	17.92	74.6
Ballast Water	36520.2	145.64	0.02	6.44	61369.4
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	24.7	180.00	5.25	6.25	5.0
Bilge Water	20.4	187.24	-2.65	1.91	133.0
Sludge	36.2	186.86	3.31	1.46	315.1
Solid cargo	20000.0	30.00	0.00	36.00	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	59509.1	104.04	0.00	16.37	66065.7
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	80380.5	106.48	-0.00	14.83	66065.7

LOADS

Location _	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HFO1	HFO Storage Tank 1	526.0	50.0	63.60	5.55	4.60	1084.6
HFO2	HFO Storage Tank 2	631.2	50.0	64.80	-5.55	4.60	1301.5
HFO3	HFO Storage Tank 3	413.4	50.0	76.71	5.40	4.65	834.7
HFO4	HFO Storage Tank 4	413.4	50.0	76.71	-5.40	4.65	834.7
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		2067.1		74.16	-0.26	4.58	4069.2
Diesel Oil density=0.900 t/m3							
T09.07	MDO Storage P	15.5	50.0	179.30	9.58	2.88	6.4
T09.08	MDO Storage S	15.5	50.0	179.30	-9.58	2.88	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.87	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.87	0.6
Total of Diesel Oil		42.2		179.80	0.00	2.88	14.0
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.1	50.0	179.80	-9.33	6.25	10.1
T10.01	Potable Water	37.8	50.0	196.40	0.00	17.06	66.2
Total of Technical Water		57.9		190.64	-3.24	13.31	76.3
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	109.6	50.0	194.82	4.84	17.92	37.3
FW2	Fresh Water S	109.6	50.0	194.82	-4.84	17.92	37.3
Total of Fresh Water		219.2		194.82	0.00	17.92	74.6

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB05.01	Water Ballast 05.01	937.4	71.0	93.54	10.58	0.93	19951.8
WB05.02	Water Ballast 05.02	937.4	71.0	93.54	-10.58	0.93	19951.8
WB05.07	Water Ballast 05.07	420.4	21.0	93.60	16.44	3.46	2085.3
WB05.08	Water Ballast 05.08	420.4	21.0	93.60	-16.44	3.46	2085.3
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.05	Water Ballast 06.05	548.7	25.4	117.38	5.58	3.63	2943.3
WB06.06	Water Ballast 06.06	548.7	25.4	117.38	-5.58	3.63	2943.3
WB06.07	Water Ballast 06.07	518.4	25.9	117.60	16.44	3.66	2085.3
WB06.08	Water Ballast 06.08	518.4	25.9	117.60	-16.44	3.66	2085.3
WB06.09	Water Ballast 06.09	603.3	100.0	117.60	5.69	11.90	0.0
WB06.10	Water Ballast 06.10	603.3	100.0	117.60	-5.69	11.90	0.0
WB06.11	Water Ballast 06.11	537.0	100.0	117.60	16.44	11.90	0.0
WB06.12	Water Ballast 06.12	537.0	100.0	117.60	-16.44	11.90	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB07.05	Water Ballast 07.05	2599.8	100.0	143.95	5.48	6.72	0.0
WB07.06	Water Ballast 07.06	2599.8	100.0	143.95	-5.48	6.72	0.0
WB07.07	Water Ballast 07.07	2390.6	100.0	143.94	16.42	6.71	0.0
WB07.08	Water Ballast 07.08	2390.6	100.0	143.94	-16.42	6.71	0.0
WB07.09	Water Ballast 07.09	715.7	100.0	143.84	5.69	11.91	0.0
WB07.10	Water Ballast 07.10	715.7	100.0	143.84	-5.69	11.91	0.0
WB07.11	Water Ballast 07.11	640.8	100.0	144.06	16.41	11.90	0.0
WB07.12	Water Ballast 07.12	640.8	100.0	144.06	-16.41	11.90	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	1199.6	100.0	167.13	15.42	7.04	0.0
WB08.08	Water Ballast 08.08	1199.6	100.0	167.13	-15.42	7.04	0.0
WB08.09	Water Ballast 08.09	351.0	100.0	168.00	5.69	12.20	0.0
WB08.10	Water Ballast 08.10	351.0	100.0	168.00	-5.69	12.20	0.0
WB08.11	Water Ballast 08.11	391.3	100.0	167.62	16.01	11.90	0.0
WB08.12	Water Ballast 08.12	391.3	100.0	167.62	-16.01	11.90	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	497.1	78.5	183.37	12.26	7.09	201.6
WB09.08	Water Ballast 09.08	497.1	78.5	183.37	-12.26	7.09	201.6
WB09.11	Water Ballast 09.11	879.9	60.0	182.71	14.67	17.77	943.5
WB09.12	Water Ballast 09.12	835.9	57.0	182.70	-14.64	17.56	921.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB10.05	Water Ballast 10.05	784.4	75.0	194.17	5.46	6.79	1249.4
WB10.06	Water Ballast 10.06	784.4	75.0	194.19	-5.43	6.79	1327.0
WB11.01	Water Ballast 11.01	795.2	48.5	205.80	0.00	4.50	2393.9
Total of Ballast Water		36520.2		145.64	0.02	6.44	61369.4
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	9.3	50.0	180.00	4.16	6.25	0.9
T09.13	Gray Water	15.4	50.0	180.00	5.91	6.25	4.1
Total of Gray Water		24.7		180.00	5.25	6.25	5.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	5.2	50.0	188.36	-0.73	3.21	0.6
T09.22	Bilge Water	15.2	50.0	186.86	-3.31	1.46	132.4
Total of Bilge Water		20.4		187.24	-2.65	1.91	133.0

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Sludge density=2.380 t/m3							
T09.21	Sludge	36.2	50.0	186.86	3.31	1.46	315.1
Solid cargo density=1.000 t/m3							
(CAS)	NON-BUOYANT CARGO	20000.0		30.00	0.00	36.00	0.0
Stores							
(STO)		100.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
Deadweight		59509.1		104.04	0.00	16.37	66065.7
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		80380.5		106.48	-0.00	14.83	66065.7

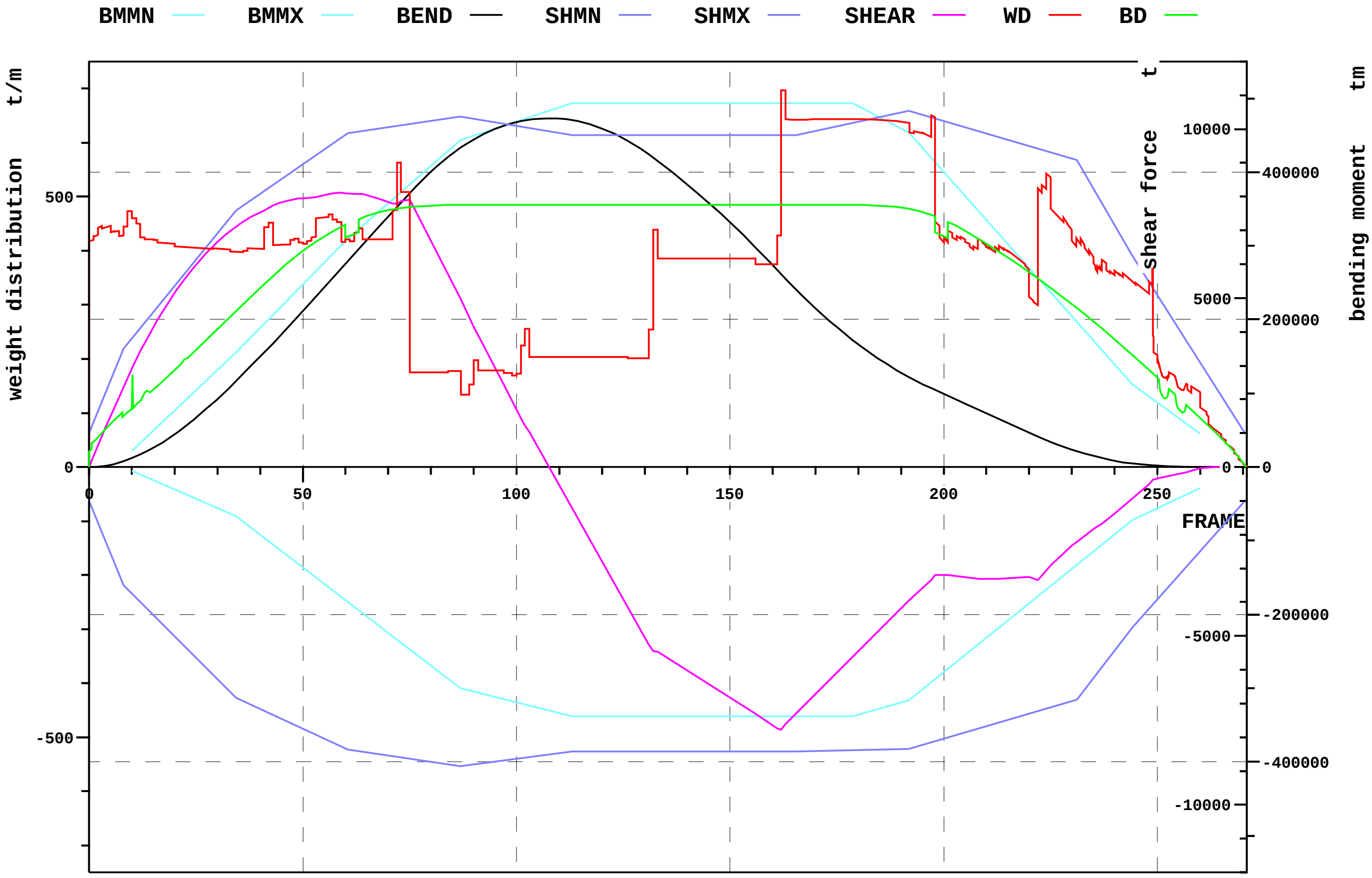


RCR	TEXT	REQ	ATTV	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.077	0.049	mrad	NOT.
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.000	mrad	NOT.
V.GZ0.2	Min. GZ > 0.2	0.200	-0.921	m	NOT.
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	7.856	deg	NOT.
V.GM0.15	GM > 0.15 m	0.150	5.466	m	OK
IMOWEATHER	IMO weather criterion	1.000	0.866		NOT.
ND-RANGE	Range of Stability acc. ND	36.000	18.578	deg	NOT.
ND-WO	Wind Overturning acc. ND	1.400	7.243		OK

Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	22.350
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	22.350
AIR_DUCT_1	UNPROTECTED	56.9	222.00	11.375	33.350	22.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

			x	frame
Shear force (min)	-7779.6 t	position:	129.6 m	162
Shear force (max)	8125.0 t		47.2 m	59
Max. rel. shear force	97.5 %		19.6 m	24
Sagging moment	-0.0 tm		216.8 m	271
Hogging moment	472949.8 tm		86.2 m	108
Max. rel. bending moment	100.0 %		77.6 m	97

LOADING CONDITION AFTLOAD2, Non-Buoyant Cargo, aft load

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELBM %
21.00	16.80	48852	-32571	82267	41.8
42.00	33.60	161610	-100351	200325	74.2
57.00	45.60	256701	-167096	288693	86.0
72.00	57.60	352765	-233865	367863	95.0
87.00	69.60	433466	-300085	443551	97.3
102.00	81.60	470089	-322267	472486	99.4
117.00	93.60	465106	-338607	493802	93.1
132.00	105.60	419656	-338607	493802	82.2
147.00	117.60	347722	-338607	493802	64.9
162.00	129.60	261456	-338607	493802	44.2
180.00	144.00	166421	-336305	489619	21.7
198.00	158.40	104186	-287350	412939	11.8
210.00	168.00	72991	-231392	334884	7.5
222.00	177.60	41326	-175443	256842	0.3
237.00	189.60	12167	-105528	159320	11.1
249.00	199.20	1991	-58776	91901	19.3

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	5332	-5135	5507	96.7
42.00	33.60	7667	-7273	8253	92.4
57.00	45.60	8101	-8146	9561	83.5
72.00	57.60	7798	-8573	10092	75.4
87.00	69.60	4965	-8852	10370	43.8
102.00	81.60	1232	-8599	10055	5.4
117.00	93.60	-2099	-8412	9823	30.8
132.00	105.60	-5441	-8412	9823	67.4
147.00	117.60	-6585	-8412	9823	80.0
162.00	129.60	-7780	-8412	9823	93.1
180.00	144.00	-5454	-8370	10215	68.6
198.00	158.40	-3202	-8104	10302	46.7
210.00	168.00	-3313	-7660	9858	50.4
222.00	177.60	-3340	-7215	9413	53.4
237.00	189.60	-1673	-5918	7808	38.1
249.00	199.20	-384	-4059	5297	21.4

3.4 SIDELOAD1

DELTAMARIN CONTRACTING LTD
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FIN-21200 RAISIO FINLAND
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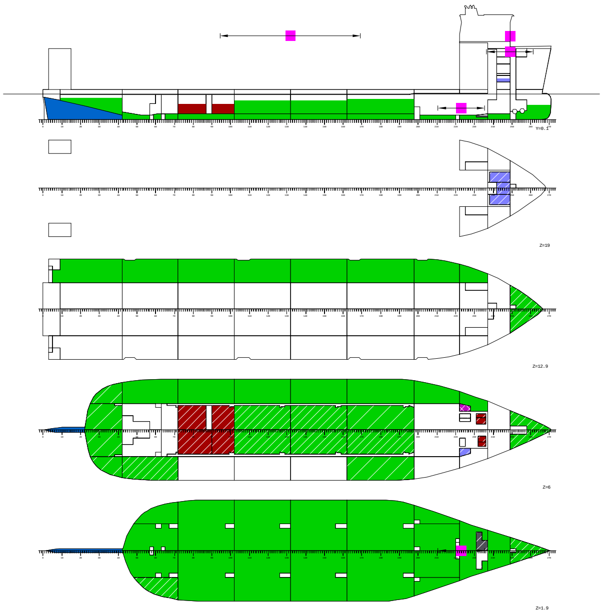


LOADING CONDITION SIDELOAD1: Non-Buoyant Cargo, half loaded from side

FLOATING POSITION

Mean draught (moulded)	11.00 m	KM above the moulded base	21.12 m
Draught at AP (moulded)	11.00 m	KG above the moulded base	10.40 m
Draught at FP (moulded)	11.00 m	GM0 (solid)	10.72 m
Trim	0.00 m	Free surface correction	-0.43 m
Heeling	0.0 deg		
		GM (fluid)	10.28 m
Mean draught (heel 0)	11.00 m		
Trim (heel 0)	0.00 m		

LOADS

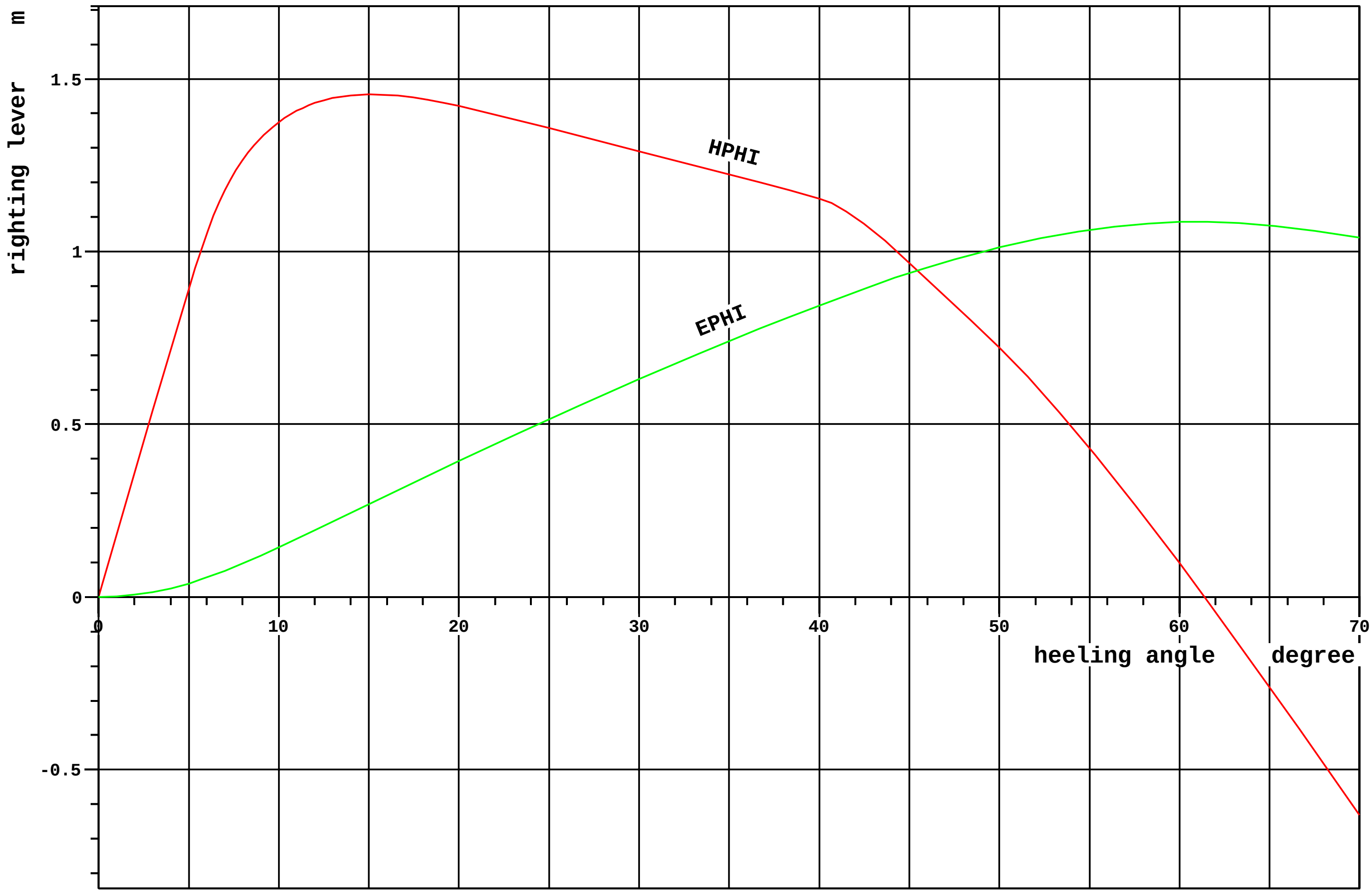
Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	2067.1	74.16	-0.26	4.58	4069.2
Diesel Oil	42.2	179.80	0.00	2.88	14.0
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	57.9	190.64	-3.24	13.31	76.3
Fresh Water	219.2	194.82	0.00	17.92	74.6
Ballast Water	46522.2	104.58	4.63	5.14	30187.1
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	24.7	180.00	5.25	6.25	5.0
Bilge Water	20.4	187.24	-2.65	1.91	133.0
Sludge	36.2	186.86	3.31	1.46	315.1
Solid cargo	10000.0	105.40	-21.50	36.00	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	59511.0	104.04	0.00	10.39	34883.4
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	80382.4	106.48	-0.00	10.40	34883.4

LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HFO1	HFO Storage Tank 1	526.0	50.0	63.60	5.55	4.60	1084.6
HFO2	HFO Storage Tank 2	631.2	50.0	64.80	-5.55	4.60	1301.5
HFO3	HFO Storage Tank 3	413.4	50.0	76.71	5.40	4.65	834.7
HFO4	HFO Storage Tank 4	413.4	50.0	76.71	-5.40	4.65	834.7
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		2067.1		74.16	-0.26	4.58	4069.2
Diesel Oil density=0.900 t/m3							
T09.07	MDO Storage P	15.5	50.0	179.30	9.58	2.88	6.4
T09.08	MDO Storage S	15.5	50.0	179.30	-9.58	2.88	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.87	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.87	0.6
Total of Diesel Oil		42.2		179.80	0.00	2.88	14.0
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.1	50.0	179.80	-9.33	6.25	10.1
T10.01	Potable Water	37.8	50.0	196.40	0.00	17.06	66.2
Total of Technical Water		57.9		190.64	-3.24	13.31	76.3
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	109.6	50.0	194.82	4.84	17.92	37.3
FW2	Fresh Water S	109.6	50.0	194.82	-4.84	17.92	37.3
Total of Fresh Water		219.2		194.82	0.00	17.92	74.6

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB02.05	Water Ballast 02.05	1143.6	75.0	23.89	5.39	7.01	2567.5
WB02.06	Water Ballast 02.06	1145.0	75.0	23.90	-5.40	7.01	2564.0
WB02.07	Water Ballast 02.07	716.0	75.2	25.22	15.54	7.74	1720.1
WB02.08	Water Ballast 02.08	666.5	70.0	25.42	-15.49	7.60	1663.4
WB02.11	Water Ballast 02.11	562.9	100.0	20.84	16.51	11.95	0.0
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	2128.4	100.0	46.50	16.11	6.30	0.0
WB03.08	Water Ballast 03.08	1383.5	65.0	46.98	-15.93	4.69	2077.9
WB03.11	Water Ballast 03.11	533.4	100.0	45.66	16.41	11.90	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.11	Water Ballast 04.11	537.0	100.0	69.60	16.44	11.90	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.05	Water Ballast 05.05	1510.2	70.0	93.49	5.39	5.46	2209.8
WB05.06	Water Ballast 05.06	1294.3	60.0	93.48	-5.43	5.04	2209.8
WB05.07	Water Ballast 05.07	2001.6	100.0	93.60	16.44	6.70	0.0
WB05.11	Water Ballast 05.11	532.8	100.0	93.66	16.40	11.90	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.05	Water Ballast 06.05	1510.2	70.0	117.49	5.39	5.46	2209.8
WB06.06	Water Ballast 06.06	1294.3	60.0	117.48	-5.43	5.04	2209.8
WB06.07	Water Ballast 06.07	2001.6	100.0	117.60	16.44	6.70	0.0
WB06.11	Water Ballast 06.11	537.0	100.0	117.60	16.44	11.90	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB07.05	Water Ballast 07.05	1945.8	74.8	143.88	5.41	5.67	3548.3
WB07.06	Water Ballast 07.06	1555.9	59.8	143.86	-5.43	5.04	2638.2
WB07.07	Water Ballast 07.07	2390.6	100.0	143.94	16.42	6.71	0.0
WB07.08	Water Ballast 07.08	1673.4	70.0	143.92	-16.41	5.48	2498.6
WB07.11	Water Ballast 07.11	640.8	100.0	144.06	16.41	11.90	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	1199.6	100.0	167.13	15.42	7.04	0.0
WB08.11	Water Ballast 08.11	391.3	100.0	167.62	16.01	11.90	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	633.4	100.0	183.26	12.72	8.16	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	623.1	38.0	205.77	0.00	3.76	2069.8
Total of Ballast Water		46522.2		104.58	4.63	5.14	30187.1
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	9.3	50.0	180.00	4.16	6.25	0.9
T09.13	Gray Water	15.4	50.0	180.00	5.91	6.25	4.1
Total of Gray Water		24.7		180.00	5.25	6.25	5.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	5.2	50.0	188.36	-0.73	3.21	0.6
T09.22	Bilge Water	15.2	50.0	186.86	-3.31	1.46	132.4
Total of Bilge Water		20.4		187.24	-2.65	1.91	133.0

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Sludge density=2.380 t/m3							
T09.21	Sludge	36.2	50.0	186.86	3.31	1.46	315.1
Solid cargo density=1.000 t/m3							
(CAS)	NON-BUOYANT CARGO	10000.0		105.40	-21.50	36.00	0.0
Stores							
(STO)		100.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
Deadweight		59511.0		104.04	0.00	10.39	34883.4
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		80382.4		106.48	-0.00	10.40	34883.4

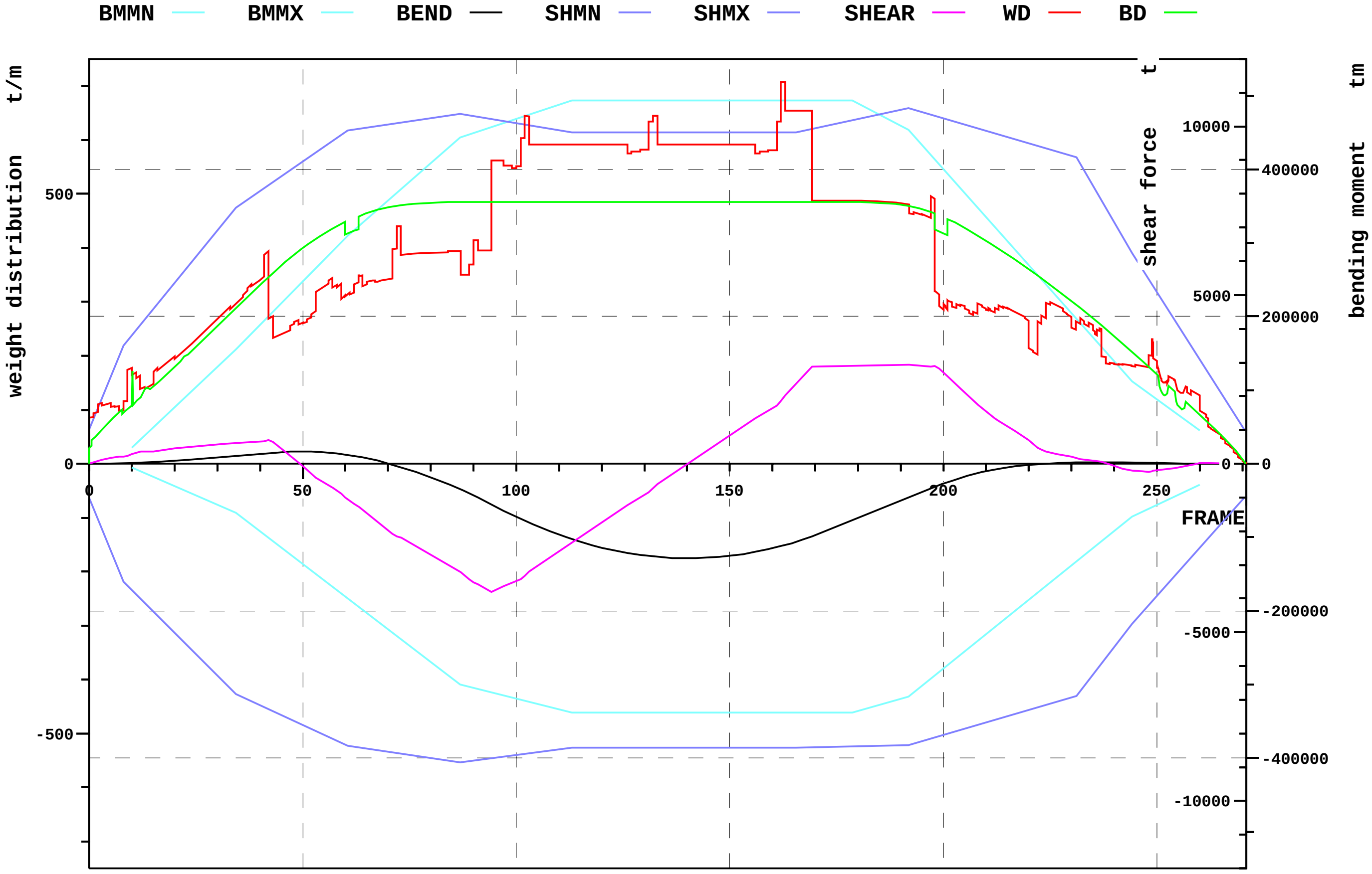


RCR	TEXT	REQ	ATTV	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.070	0.271	mrad	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.213	mrad	OK
V.GZ0.2	Min. GZ > 0.2	0.200	1.290	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	15.133	deg	OK
V.GM0.15	GM > 0.15 m	0.150	10.285	m	OK
IMOWEATHER	IMO weather criterion	1.000	2.484		OK
ND-RANGE	Range of Stability acc. ND	36.000	61.375	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	24.315		OK

Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	22.350
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	22.350
AIR_DUCT_1	UNPROTECTED	56.8	222.00	11.375	33.350	22.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

			x	frame
Shear force (min)	-3806.5 t	position:	75.4 m	94
Shear force (max)	2925.4 t		153.6 m	192
Max. rel. shear force	48.0 %		75.4 m	94
Sagging moment	-128317 tm		112.8 m	141
Hogging moment	16526.4 tm		39.6 m	50
Max. rel. bending moment	55.3 %		8.0 m	10

LOADING CONDITION SIDELOAD1, Non-Buoyant Cargo, half loaded from side

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELBM %
21.00	16.80	4661	-32571	82267	35.2
42.00	33.60	14372	-100351	200325	23.7
57.00	45.60	14106	-167096	288693	20.5
72.00	57.60	-3051	-233865	367863	23.3
87.00	69.60	-35014	-300085	443551	28.7
102.00	81.60	-77724	-322267	472486	38.5
117.00	93.60	-109524	-338607	493802	45.0
132.00	105.60	-126050	-338607	493802	48.9
147.00	117.60	-126742	-338607	493802	49.1
162.00	129.60	-112184	-338607	493802	45.6
180.00	144.00	-73568	-336305	489619	36.4
198.00	158.40	-31690	-287350	412939	27.0
210.00	168.00	-10285	-231392	334884	21.9
222.00	177.60	-464	-175443	256842	19.0
237.00	189.60	2175	-105528	159320	18.7
249.00	199.20	838	-58776	91901	20.9

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	473	-5135	5507	5.4
42.00	33.60	703	-7273	8253	2.7
57.00	45.60	-717	-8146	9561	16.1
72.00	57.60	-2152	-8573	10092	31.2
87.00	69.60	-3211	-8852	10370	41.3
102.00	81.60	-3330	-8599	10055	43.5
117.00	93.60	-2007	-8412	9823	29.8
132.00	105.60	-739	-8412	9823	15.8
147.00	117.60	585	-8412	9823	1.3
162.00	129.60	1850	-8412	9823	12.6
180.00	144.00	2898	-8370	10215	21.3
198.00	158.40	2906	-8104	10302	19.6
210.00	168.00	1553	-7660	9858	5.2
222.00	177.60	476	-7215	9413	7.5
237.00	189.60	67	-5918	7808	12.8
249.00	199.20	-213	-4059	5297	17.8